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WO-500-04 — Replace VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3)

Area	A500 Pipe rack	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	rack-bent:720 — Pipe rack bent at E 60'-0" — E 60'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	1"-P150-HS-5011
Isometrics	ISO-HS-5011 (owner's drawings, attached)		
Permits	General work permit		

Reason

Hose-station take-off valve at the water header seized.

Scope of work

1. Isolate per the isolation list, drain and vent 1"-P150-HS-5011 between the isolations.
2. Unbolt and remove VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3) at E 62'-6" N -4'-0" EL 123'-7-1/2".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, BALL, FLOATING 1" P150 of the same size, class and face-to-face (BALL1-P150), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 1"-P150-HS-5011/A3, 1"-P150-HS-5011/A2, 1"-P150-HS-5011/A5

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-HS-5011	1"-P150-HS-5011	1"-P150-HS-5011	1	agrees with the model

ISO-HS-5011 (PDF): ../isometrics-imported/ISO-HS-5011.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:720	structure	Pipe rack bent at E 60'-0"		E 60'-0" N 0.000" EL 122'-0"
1"-P150-HS-5011/A3	item	VALVE, BALL, FLOATING 1" P150	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 123'-7-1/2"
1"-P150-HS-5011/A2	item	GASKET 1" P150 COMPRESSED SHEET	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 123'-3/4"
1"-P150-HS-5011/A5	item	GASKET 1" P150 COMPRESSED SHEET	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 124'-2-1/4"
3"-P150-W-2008/T:1"-P150-HS-5011	item	TEE REDUCING 3" × 1" P150	3"-P150-W-2008	E 62'-6" N -4'-0" EL 122'-6"
1"-P150-HS-5011/B3	item	VALVE, BALL, FLOATING 1" P150	1"-P150-HS-5011	E 62'-6" N 11'-0" EL 104'-2-1/8"
1"-P150-HS-5011/B6	item	FLANGE WELD-NECK 1" P150 RAISED FACE	1"-P150-HS-5011	E 62'-6" N 11'-0" EL 104'-11-1/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	3"-P150-W-2008/T:1"-P150-HS-5011	3"-P150-W-2008	No valve on this side within the line: isolate or depressure the parent line 3"-P150-W-2008 at its tee under the shutdown isolation			
2	valve	1"-P150-HS-5011/B3	1"-P150-HS-5011	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/B3)	E 62'-6" N 11'-0" EL 104'-2-1/8"		
3	blind	1"-P150-HS-5011/B6	1"-P150-HS-5011	Fit a line blind between 1"-P150-HS-5011/B3 and FLANGE WELD-NECK 1" P150 RAISED FACE (1"-P150-HS-5011/B6), work side	E 62'-6" N 11'-0" EL 104'-8-15/16"		

Parts

Description	Qty	Replaces
VALVE, BALL, FLOATING 1" P150	1	1"-P150-HS-5011/A3
GASKET 1" P150 COMPRESSED SHEET	1	1"-P150-HS-5011/A2
STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	1	1"-P150-HS-5011/A4
GASKET 1" P150 COMPRESSED SHEET	1	1"-P150-HS-5011/A5

Description	Qty	Replaces
STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	1	1"-P150-HS-5011/A7

Inspections

Id	Inspection	Criteria	Basis
WO-500-04-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
1"-P150-HS-5011/A2	4 × 0.5" studs (1"-P150-HS-5011/A4)	8,836	3	
1"-P150-HS-5011/A5	4 × 0.5" studs (1"-P150-HS-5011/A7)	8,836	3	

Test — Service leak test

Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

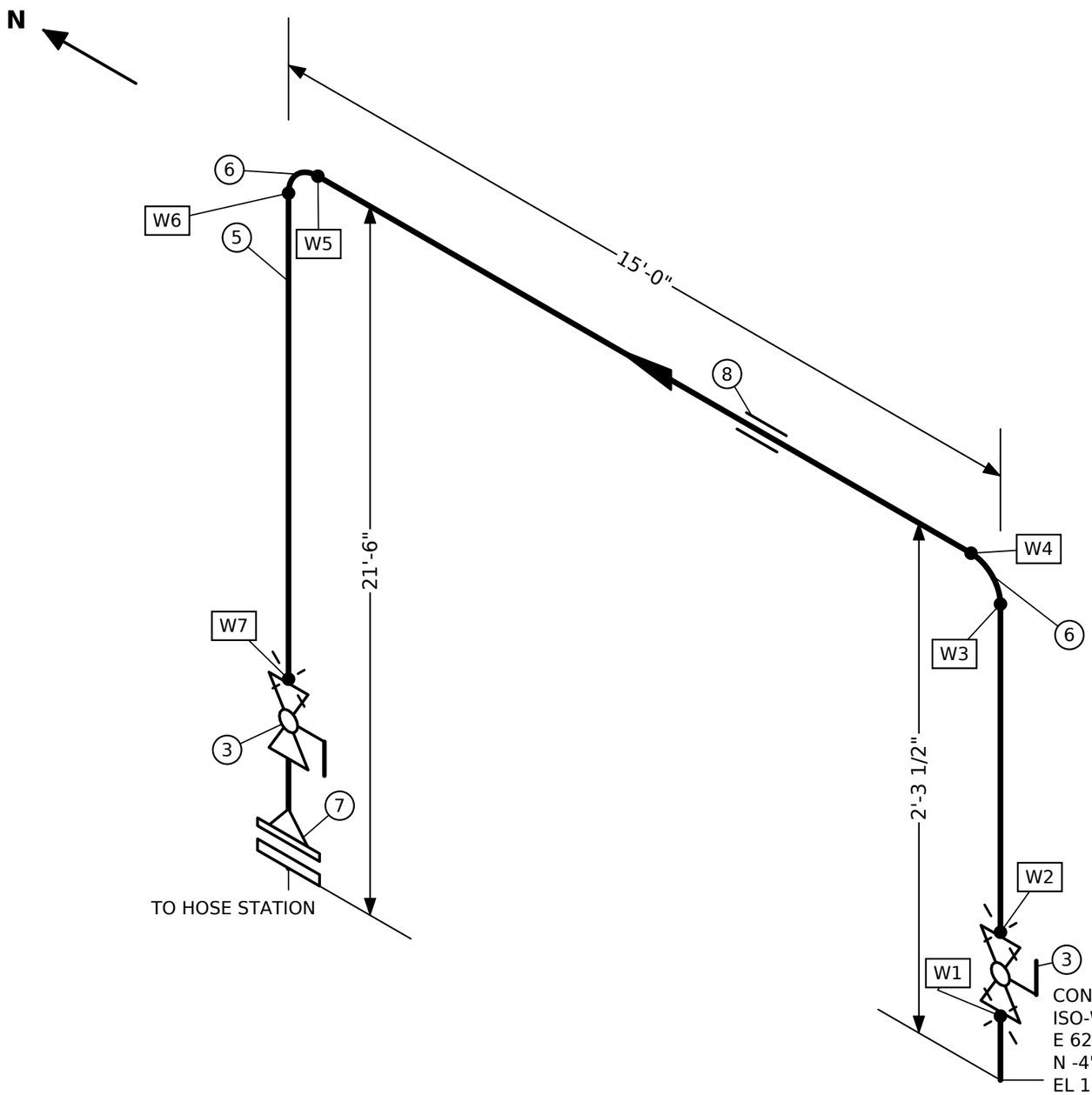
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 1"-P150-HS-5011/A3 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	BALL1-P150	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
1"-P150-HS-5011/A3	affected	ISO-HS-5011	1	
1"-P150-HS-5011/A2	affected	ISO-HS-5011	1	
1"-P150-HS-5011/A5	affected	ISO-HS-5011	1	
3"-P150-W-2008/T:1"-P150-HS-5011	isolation: equipment	ISO-W-2008	3	
1"-P150-HS-5011/B3	isolation: valve	ISO-HS-5011	1	
1"-P150-HS-5011/B6	isolation: blind	ISO-HS-5011	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	1"	FLANGE WELD-NECK 1" P150 RAISED FACE
2	4	1"	GASKET 1" P150 COMPRESSED SHEET
3	2	1"	VALVE, BALL, FLOATING 1" P150
4	4	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
5	34'-9 11/16"	1"	PIPE 1" P150 BEVELLED ENDS
6	2	1"	ELBOW 90° R = 5/3 OD 1" P150
7	1	1"	FLANGE BLIND 1" P150 RAISED FACE
8	1	1"	HANGER FOR 1" LINE (HANGER-1x98.0)

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

WATER (COOLING AND PROCESS)

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	HS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-HS-5011	W150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-HS-5011	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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